

## AGADEZ, Niger

WMO: 610240

Lat: 16.967N

Lon: 7.983E

Elev: 1657

StdP: 13.84

Time Zone: 1.00 (E01)

Period: 82-15

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 51.3       | 53.8 | 17.2                          | 13.9 | 75.8 | 17.2 | 13.9 | 75.8 | 36.1                  | 68.8 | 33.4 | 71.2 | 6.9                   | 100  | 0.539 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |       |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|-------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%    |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB    | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)   | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 5             | 22.7                   | 108.0           | 66.8 | 107.0 | 66.7 | 105.7 | 66.6 | 75.5                | 92.5 | 74.6 | 91.9 | 73.8 | 91.5 | 10.2                 | 140  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 71.5                            | 123.7 | 81.6 | 70.0 | 117.6 | 82.2 | 68.7 | 112.3 | 83.0 | 40.1          | 92.1 | 39.3 | 91.6 | 38.5 | 91.2 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (d)                        | (e)  | (f)                | (g) | (h)                                                | (i)  | (j)        | (k)  | (l)        | (m)  | (n)        | (o)  | (p)  | (q) |
| (4) 30.0          | 25.6 | 22.5 | DB                         | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  | N/A  | N/A |
| (5)               |      |      | WB                         | 34.9 | 79.7               | 3.4 | 3.2                                                | 32.5 | 81.9       | 30.6 | 83.8       | 28.7 | 85.6       | 26.2 | 87.9 |     |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 83.9   | 68.5  | 74.4  | 81.6  | 89.5  | 94.1  | 94.3  | 91.3  | 88.6  | 90.5  | 85.7  | 77.8  | 70.8  |
|                                                                        | DBStd       | 9.85   | 5.41  | 6.88  | 6.44  | 5.09  | 3.40  | 2.66  | 3.30  | 3.21  | 2.25  | 4.07  | 4.65  | 5.35  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 38     | 24    | 5     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 9     |
|                                                                        | CDD50       | 12390  | 572   | 684   | 978   | 1184  | 1367  | 1329  | 1279  | 1197  | 1215  | 1108  | 833   | 644   |
|                                                                        | CDD65       | 6955   | 132   | 269   | 513   | 734   | 902   | 879   | 814   | 732   | 765   | 643   | 384   | 188   |
|                                                                        | CDH74       | 103305 | 1377  | 3320  | 7090  | 11727 | 15195 | 14638 | 12453 | 10280 | 11750 | 9119  | 4407  | 1949  |
| (13)                                                                   | CDH80       | 65522  | 458   | 1639  | 4116  | 7922  | 10812 | 10346 | 8101  | 6115  | 7548  | 5535  | 2215  | 715   |
| (14)                                                                   | Wind        | WSAvg  | 10.1  | 12.1  | 10.9  | 11.5  | 10.5  | 9.4   | 9.3   | 9.2   | 8.4   | 9.8   | 9.8   | 10.8  |
| (15) Precipitation                                                     | PrecAvg     | 4.6    | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 0.4   | 1.7   | 2.2   | 0.6   | 0.0   | 0.0   | 0.0   |
|                                                                        | PrecMax     | 8.9    | 0.0   | 0.0   | 0.2   | 2.0   | 1.3   | 1.5   | 4.9   | 5.9   | 3.1   | 0.2   | 0.0   | 0.0   |
|                                                                        | PrecMin     | 0.2    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
|                                                                        | PrecStd     | 1.9    | 0.0   | 0.0   | 0.0   | 0.3   | 0.4   | 0.4   | 1.1   | 1.3   | 0.7   | 0.0   | 0.0   | 0.0   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 92.9  | 100.3 | 104.2 | 108.3 | 109.9 | 109.5 | 107.9 | 104.6 | 105.7 | 103.8 | 98.9  | 93.6  |
|                                                                        |             | MCWB   | 58.6  | 62.1  | 64.6  | 67.0  | 66.9  | 66.9  | 68.7  | 67.4  | 66.5  | 65.3  | 63.0  | 61.0  |
|                                                                        | 2%          | DB     | 88.8  | 96.6  | 101.7 | 106.4 | 108.6 | 107.7 | 105.7 | 102.9 | 104.0 | 101.9 | 96.7  | 90.1  |
|                                                                        |             | MCWB   | 57.0  | 61.0  | 63.5  | 65.6  | 66.6  | 67.0  | 68.7  | 67.9  | 66.4  | 64.9  | 61.7  | 58.9  |
|                                                                        | 5%          | DB     | 85.1  | 93.1  | 99.0  | 104.9 | 107.3 | 106.3 | 103.7 | 101.1 | 102.5 | 100.3 | 93.6  | 86.8  |
|                                                                        |             | MCWB   | 55.8  | 59.2  | 62.2  | 65.2  | 66.3  | 67.2  | 68.5  | 68.3  | 66.6  | 63.8  | 60.2  | 57.4  |
|                                                                        | 10%         | DB     | 81.4  | 89.4  | 96.4  | 103.1 | 105.7 | 105.0 | 101.5 | 98.7  | 101.0 | 98.4  | 91.2  | 83.9  |
|                                                                        |             | MCWB   | 54.2  | 57.2  | 60.8  | 64.5  | 66.1  | 67.1  | 69.2  | 69.9  | 66.7  | 63.1  | 59.6  | 56.1  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 60.6  | 65.2  | 68.2  | 70.8  | 75.2  | 75.9  | 76.6  | 77.0  | 75.6  | 72.7  | 64.9  | 63.7  |
|                                                                        |             | MCDB   | 87.3  | 95.1  | 96.4  | 99.6  | 100.0 | 98.0  | 93.4  | 91.3  | 93.5  | 92.1  | 94.2  | 88.1  |
|                                                                        | 2%          | WB     | 58.0  | 62.2  | 65.6  | 68.9  | 73.3  | 74.3  | 75.1  | 75.5  | 74.0  | 69.6  | 62.8  | 60.4  |
|                                                                        |             | MCDB   | 86.0  | 92.8  | 95.5  | 100.9 | 98.4  | 95.9  | 93.0  | 89.7  | 92.0  | 93.0  | 93.4  | 87.7  |
|                                                                        | 5%          | WB     | 56.2  | 59.9  | 63.5  | 67.1  | 71.6  | 73.0  | 74.1  | 74.6  | 72.6  | 67.2  | 61.4  | 58.3  |
|                                                                        |             | MCDB   | 83.9  | 91.0  | 95.9  | 101.1 | 96.3  | 95.5  | 92.2  | 89.1  | 90.9  | 94.8  | 91.8  | 85.5  |
|                                                                        | 10%         | WB     | 54.5  | 57.9  | 61.6  | 65.5  | 70.0  | 71.9  | 73.2  | 73.8  | 71.5  | 64.8  | 60.1  | 56.3  |
|                                                                        |             | MCDB   | 80.8  | 88.4  | 94.1  | 100.2 | 94.7  | 95.3  | 91.5  | 88.7  | 90.3  | 95.0  | 89.9  | 82.6  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 25.0  | 26.4  | 25.5  | 25.3  | 22.7  | 21.9  | 20.6  | 20.3  | 21.9  | 23.9  | 25.4  | 25.3  |
|                                                                        |             | MCDBR  | 27.9  | 28.2  | 26.3  | 25.2  | 23.9  | 23.0  | 22.6  | 22.6  | 23.6  | 24.2  | 27.1  | 27.7  |
|                                                                        |             | MCWBR  | 13.1  | 12.3  | 11.4  | 10.8  | 9.5   | 9.4   | 8.3   | 7.2   | 8.9   | 9.5   | 11.6  | 12.6  |
|                                                                        |             | MCDBR  | 27.6  | 27.4  | 24.4  | 24.1  | 20.8  | 21.0  | 20.2  | 19.6  | 20.8  | 22.4  | 26.5  | 26.9  |
|                                                                        | 5% WB       | MCWBR  | 13.3  | 12.4  | 11.1  | 11.1  | 9.3   | 9.3   | 7.2   | 6.4   | 8.8   | 10.6  | 11.9  | 12.8  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.466 | 0.521 | 0.594 | 0.669 | 0.714 | 0.731 | 0.715 | 0.684 | 0.654 | 0.603 | 0.498 | 0.461 |
|                                                                        | taud        |        | 1.914 | 1.760 | 1.611 | 1.510 | 1.445 | 1.451 | 1.535 | 1.592 | 1.598 | 1.637 | 1.806 | 1.895 |
|                                                                        | Ebn at Noon |        | 257   | 249   | 236   | 220   | 207   | 201   | 205   | 213   | 219   | 224   | 245   | 253   |
|                                                                        | Edh at Noon |        | 58    | 71    | 85    | 95    | 100   | 98    | 90    | 86    | 85    | 80    | 64    | 58    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1724  | 1912  | 2141  | 2240  | 2244  | 2159  | 2153  | 2065  | 2048  | 1946  | 1791  | 1650  |
| (45)                                                                   | RadStd      |        | 42    | 73    | 98    | 87    | 66    | 96    | 50    | 48    | 54    | 58    | 40    | 52    |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (0 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page